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Education

Dartmouth College, Hanover, NH Sep 2019 – June 2024
- Ph.D., Mathematics
- Thesis Advisor: Sergi Elizalde
Lafayette College, Easton, PA Sep 2015 – May 2019
- B.S. Mathematics with Honors
- Minor Computer Science

Professional Experience

Howard University, Washington, DC September 2026 – Present
- Temporary Lecturer, Department of Mathematics
Villanova University, Villanova, PA August 2025 – May 2026
- Visiting Assistant Teaching Professor, Department of Mathematics and Statistics
Florida Gulf Coast University, Fort Myers, FL August 2024 – May 2025
- Visiting Assistant Professor, Department of Mathematics

Papers

Publications

4. *Counting Boolean intervals in the weak Bruhat order of a finite Coxeter group* (with J. Elder, P. E. Harris, J. C. Martínez Mori) <https://doi.org/10.47443/dml.2024.163> *Discrete Math. Lett.* 15 (2025) 52-59
3. *On the Order of P-Strict Promotion on $V \times [\ell]$* *Electron. J. Combin.* 32(2) (2025) #P2.8 <https://doi.org/10.37236/13044>
2. *Coloring Groups* (with A. Wilson) *Discrete Math & Theoretical Computer Science* vol. 26:2 (2024) <https://doi.org/10.46298/dmtcs.12753>
1. *Rowmotion on 321-avoiding permutations* (with S. Elizalde) *Electron. J. Combin.* 30(3) (2023) #P3.5 <https://doi.org/10.37236/11792>

Preprints and In Preparation

- *Dimensions of toggleability spaces* (with S. Daugherty, N. Mayers) <https://doi.org/10.48550/arXiv.2508.14974> (submitted)

- *The poset of maximal tubings of the cycle graph is a lattice* (with E. Barnard, M. Hlavacek, B. Kagy, N. Lesnevich, G. D. Nasr, K. Waddle) <https://doi.org/10.48550/arXiv.2510.09429> (submitted)
- *Enumerating Pattern Avoiding Parking Functions* <https://doi.org/10.48550/arXiv.2605.24092>
- *Parking Cascades: From the simplest sequence to Motzkin and Catalan* (with J. Elder, P. E. Harris, M. Lynch, J. C. Martínez Mori) <https://doi.org/10.48550/arXiv.2608.00735> (submitted)
- *Invariance of rowmotion for variants of the Tamari lattice* (with E. Barnard, C. Ceballos C. Chenevière, C. Defant, S. Hopkins, M. Müller, M. Rubey, J. Striker) <https://doi.org/10.48550/arXiv.2609.12983>
- *A Distributive Lattice on 132-Avoiding Alternating Sign Matrices* (In Preparation)

Presentations

Invited Conference Presentations

- “Independence Polytopes: a Novel GPT Application” *AMS Special Session on Posets, lattices, and semilattices in algebraic and geometric combinatorics* at Joint Math Meetings January 2027
- “On the q -Poincaré sum of a Coxeter group” *AMS Special Session on Algebraic and Enumerative Combinatorics* at AMS Spring Eastern Sectional October 2026
- “The poset of maximal tubings of the cycle graph” *AMS Special Session on Dynamical Algebraic Combinatorics* at AMS Spring Central Sectional April 2026
- “Dimensions of toggleability spaces” *AMS Special Session on Algebraic Combinatorics* at AMS Spring Southeastern Sectional March 2026
- “Pattern avoiding Parking Functions” *Special Session on Permutation Patterns* at Joint Math Meetings January 2026
- “On the Toggleability Spaces of Diagrams” *AMS Special Session on Dynamical Algebraic Combinatorics* at AMS Spring Eastern Sectional April 2025
- “On the Order of P-Strict Promotion on $V \times [\ell]$ ” *AMS Special Session on Enumerative and Dynamical Combinatorics* at AMS Spring Southeastern Sectional March 2025
- “Avoiding a Single Pattern in the Bruhat Order”, *AMS Special Session on Permutation Patterns* at AMS Fall Eastern Sectional October 2024
- “Rowmotion on 321-Avoiding Permutations”, *AMS Special Session on Algebraic and Bijective Methods in Permutation Enumeration* at Joint Math Meetings April 2022

Contributed Research Presentations

- “Dimensions of toggleability spaces” (Poster), *Formal Power Series and Algebraic Combinatorics* July 2026
- “Avoiding a Single Pattern in the Bruhat Order” (Poster), *Permutation Patterns* July 2025
- “Boolean intervals in the weak Bruhat order of a finite Coxeter group” *Graduate Student Combinatorics Conference* March 2024
- “Coloring Groups” (Poster), *Combinatorial Algebra meets Algebraic Combinatorics* January 2024
- “Rowmotion on 321-Avoiding Permutations” (Poster), *Formal Power Series and Algebraic Combinatorics* July 2023

- “Rowmotion on 321-Avoiding Permutations”, *Permutation Patterns* July 2023

Invited Seminar Talks

Dartmouth College	May 2026
Lafayette College	April 2026
George Washington University	March 2026
Drexel University (CAGE seminar)	March 2026
Drexel University (CAGE seminar)	May 2025
University of Florida	April 2025
Kennesaw State University	April 2025
University of Wisconsin – Milwaukee	April 2024
Graduate Online Combinatorics Colloquium	November 2023
LACIM at Université du Québec à Montréal	November 2023
Michigan State University	November 2023
North Dakota State University	November 2023
Dartmouth College	May 2022
University of Vermont	April 2022

Other Conferences and Workshops Attended

- Webs in Algebra, Geometry, Topology and Combinatorics at ICERM	December 2025
- BIRS 5-Day Workshop on Lattice Theory	January 2025
- Collaborate @ ICERM	January 2025
- AMS MRC in Algebraic Combinatorics	June 2024
- Richard Stanley 80th Birthday Conference	June 2024
- ALCOVR Workshop in Dynamical Algebraic Combinatorics	May 2024
- AMS Spring Eastern Sectional	April 2024
- 48th Annual New York State Regional Graduate Mathematics Conference	April 2023
- Permutation Patterns	June 2022
- BIRS 5-Day Workshop on Dynamical Algebraic Combinatorics (Virtual)	November 2021
- Permutation Patterns (Virtual)	April 2021

Teaching Experience

Instructor *Howard University*

- Math 006 College Algebra	Fall 2026
- Math 156 Calculus 1	Fall 2026
- Math 159 Differential Equations	Fall 2026

Instructor *Villanova University*

- MAT 1505 Calculus 2	Spring 2026
- MAT 2500 Calculus 3 (2 Sections)	Spring 2026

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- MAT 1320 Calculus 1 for Liberal Arts Majors Fall 2025
- MAT 1500 Calculus 1 (2 Sections) Fall 2025

Instructor *Florida Gulf Coast University*

- MAC 2233 Elementary Calculus 1 (2 Sections) Spring 2025
- MAS 3105 Linear Algebra Spring 2025
- MAT 1033 Intermediate Algebra (2 Sections) Fall 2024
- MAC 1147 Precalculus Fall 2024

Instructor *Dartmouth College*

- Math 28 Introduction to Combinatorics Winter 2024
- Math 9 Multivariable Differential Calculus with Linear Algebra Fall 2022
- Math 3 Introduction to Calculus Fall 2021

Teaching Assistant, *Dartmouth College*

- Math 28 Introduction to Combinatorics Winter 2023
- Math 3 Introduction to Calculus Winter 2021 online over Zoom
- Math 108 Topics in Combinatorics Fall 2020 online over Zoom
- Math 10 Introductory Statistics Spring 2020 online over Zoom
- Math 8 Calculus of Functions of one and Several Variables Fall 2019

Service and Outreach

- Dartmouth Combinatorics Seminar Co-organizer Fall 2022 – Spring 2024
- Dartmouth College *Directed Reading Program* Co-organizer Winter 2023 – Winter 2024
- Florida Gulf Coast University Colloquium Committee August 2024 – May 2025
- Florida Gulf Coast University ASPIRE Committee August 2024 – May 2025

Referee For

Annals of Combinatorics, Discrete Math & Theoretical Computer Science, Electronic Journal of Combinatorics (x3), European Journal of Combinatorics, Journal of Integer Sequences